Emanuel André Medina Araujo

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Education

Technical University of Munich — Oct. 2026 – Present
Master of Science in Informatics — *Munich, DE*

Technical University of Munich — Oct. 2020 – Aug. 2026
Bachelor of Science in Informatics (Computer Science), Application Area in Physics — *Munich, DE*

Technical University of Munich — Oct. 2024 – Sep. 2025
Parallel studies in Physics (B.Sc. programme, not completed) — *Munich, DE*

Herzog-Christoph-Gymnasium — Sep. 2012 – Sep. 2020
Abitur, Awarded MINT-EC certificate with distinction — *Beilstein, DE*

Experience

Working Student Software Engineer — Nov. 2024 – Present
Vector Informatik GmbH - vSceneCreator (3D OpenDRIVE Editor) — *Munich, DE*

- Helping take vSceneCreator (Unity/C#) from its first alpha (0.1) to the stable 1.0 release (Dec. 2026), with features suggested, designed and implemented in close collaboration with Product Management.
- Reduced edit-mode switch times by up to 70% (previously about 2 min for a 40 MB file) by restructuring the editor's base model to lazy-load non-essential and non-visible objects, creating editing handles on selection instead of up front.
- Co-developed incremental object creation that sped up file loading by 10%, replacing application freezes with an enhanced progress bar.
- Cut CI time for editor-only pull requests from 29.8 to 18.8 min and from 43.9 to 33.7 min on two build nodes (Jenkins) by skipping unrelated packaging, reusing module results within a run, compressing archives about 5x faster, and retrying flaky tests in place.
- Built a polyline trajectory editing mode with load and save for OpenSCENARIO (driving-scenario standard): data model, controllers, inspectors, validation issues, and a lossless round-trip of optional XML attributes, so absent values are no longer written as 0.
- Designed and implemented geometry simplification that reduces OpenDRIVE road-network complexity by detecting consecutive line and arc segments with matching heading and curvature and merging them through single or bulk fixes, with undo/redo, live validation and PlayMode tests.
- Developed a read-only Explore Mode as the default for users without a licence, with a hierarchical overview tree of the whole network and a hover readout of world and road (s/t) coordinates, keeping inspection separate from editing features; covered by a 1,200-line PlayMode test suite.
- Extended the Explore Mode readout to every edit mode through a central collider manager, and added a context menu that copies OpenSCENARIO position snippets for the hovered point.
- Implemented support for single-sided lanes in OpenDRIVE import by automatically converting them to valid two-sided sections and correcting lane links when sections split, preventing broken files; verified with file round-trip tests.
- Migrated the editor's geometry and road-network code across 108 files from custom double-precision vector types to Unity.Mathematics equivalents, with extension utilities and adapted tests.
- Brought the entire vSceneCreator application to Linux: abstracted platform-specific dialog and file-operation handling, wrote C/C++ for the licence check, and set up Conan to install Windows and Linux plugins.
- Streamlined developer tools by improving the Unity Rider integration, resulting in 10% faster asset refresh times.
- Helped set up the team's AI-assistant configuration and initiated an autonomous review-agent loop.
- Added a development auto-save with per-session folders, 30-day cleanup and a 500 MB session cap, and made user settings persist only on change using C# records.
- Improved keyboard navigation with shortcuts that adapt per mode, shortcut tooltips and keyboard cycling of dropdowns; visualised the road boundary in plan view.
- Strengthened geometry test coverage with road-mark mesh tests against reference meshes, lane-height mesh assertions with geometric helpers, and a shared geometry-element fixture.
- Introduced automated formatting across 1,202 files, with editorconfig rules, a clean-up profile and blame-ignore revisions, and wired project git hooks into the developer setup.
- Scoped own work by writing 67 tickets, including feature specifications, and reviewed teammates' pull requests.

Working Student Software Engineer Mar. 2021 – Oct. 2024

Vector Informatik GmbH - DYNA4 (Simulation environment for virtual driving tests) *Munich, DE*

- Developed features, fixed critical bugs and enhanced the UI across nine releases (6.0 to 9.1) in Java and Eclipse RCP, in direct collaboration with Product Management.
- Engineered a custom editor and new file format for signal aliases and favourites, with a persistence layer, database sync and project migration, using built-in validation and quick fixes so users need less prior knowledge.
- Built reference analysis so users save only the files a database sync actually references, replacing prompts for every unsaved file.
- Co-developed warning markers and a save quick fix for simulations with unsaved parameter files, removing a deadlock risk by no longer querying the UI from background workspace jobs.
- Unified simulation-node messages and warnings across the Problems, Tasks and Simulation views and the parameter dialog, and added problem markers with quick fixes for common project and simulation errors.
- Extended the standalone run-package export from scenarios to tasks, refactoring the simulation runner to export instead of simulate, with SWTBot UI tests of the wizard and output structure.
- Co-developed automatic project migration: incompatible projects are detected and closed at startup and migrated in a background job by a new migration service, with a project backup step and JUnit plugin tests.
- Fixed internal errors when renaming the active project by replacing custom rename and delete actions with Eclipse refactoring participants.
- Created a guided clean-up dialog that finds and removes unreferenced project files through a new file-reference service, and refactored a dialog to show all unsaved artifacts.
- Developed OpenSCENARIO tooling (driving-scenario standard): a New File wizard (scenario, catalog, variation) creating files via EMF, rule-based icon mapping, directory-reference analysis for clean-up, and a cache-key fix for deprecated elements.
- Evolved data models and formats: EMF parameter-model changes, XML schema generation for package descriptors, and diff-friendly project properties files for version control.
- Reworked parameter-variation dialogs for design-of-experiments workflows, adding named values, direct editing of value arrays and multi-file selection, and surfaced varied parameters in the results.
- Enhanced usability by reworking the trace signal view with full signal paths, an extended filter box, search across comments and tooltips, double-click signal adding, source-tree highlighting and a prominent refresh hint.
- Added a multi-axis signal plot that groups signals by physical unit (JavaFX) and toolbar actions to plot selected data in MATLAB or the built-in view.
- Reduced manual QA effort with SWTBot UI tests, JUnit tests and a custom JUnit formatter that links Jenkins results to Jira tickets through Xray.
- Wrote reusable SWTBot UI-test wrappers and matchers for an in-house form framework, plus Simulation-view tests that drive MATLAB.
- Cleared IDE problems across 861 files, removed deprecated APIs and applied save actions across 1,283 files, stopping formatting noise in later pull requests.
- Excluded irrelevant files from unsaved-file prompts, fixed triple save requests, and automated a dependency path in the version-bump Python script.

Projects

LoveNeverDies | *Legal Loves Tech Hackathon – Bavarian Ministry of Justice Challenge* Aug. 2026

- Built an end-to-end encrypted estate assistant for bereaved families (React, TypeScript, Supabase) in ten days with a team of lawyers, guiding users through inheritance deadlines via a 141-node legal decision tree.
- Designed a zero-trust architecture in which the server only stores ciphertext: hybrid post-quantum key exchange and signatures (ML-KEM + X25519, ML-DSA + Ed25519), AES-256-GCM, and a Shamir-secret-shared vault.
- Verified row-level security with 130 Vitest files against the real Postgres migrations (PGlite/WASM) and multi-user Playwright end-to-end tests; enforced the layered architecture through ESLint rules.

SchafPlay | *Peer-to-peer multiplayer card game (PWA)* Jun. 2026 – Aug. 2026

- Built an offline-first PWA for the Bavarian card game Schafkopf (React, TypeScript) with serverless peer-to-peer multiplayer over WebRTC data channels, using a manual SDP exchange instead of a signaling server.
- Implemented a host-authoritative game engine with per-player state redaction, so opponents' hands never go over the wire, plus heuristic AI opponents and a trick-by-trick replay analysis.
- Covered pairing, reconnect, and rule enforcement with a seeded Playwright end-to-end suite in GitHub Actions CI.

Autonomous AI Agent Orchestration (Hermes) | *Headless Server Infrastructure & Security* May 2026 – Present

- Architected and deployed a headless orchestration server on repurposed hardware running Ubuntu Server Pro, enforcing strict privacy via LUKS full-disk encryption, SSH key-based authentication, and Tailscale network isolation.

- Configured and launched the Hermes autonomous AI agent (Nous Research) as a personal assistant for everyday organization and tasks, leveraging its native Bitwarden Secrets Manager integration for secure, centralized credential handling.
- Engineered a real-time secure messaging gateway utilizing `signal-cli` wrapped in systemd services and D-Bus for agent notifications.

6D Object Pose Estimation for Robotics | *Robotics Seminar Paper* Apr. 2026 – Jul. 2026

- Wrote an IEEE-format survey of 6D object pose estimation, from classical geometric registration to zero-shot foundation-model methods, evaluated against generalization, real-time capability, and accuracy.
- Showed that no current method meets all three requirements and identified hybrid architectures (slow zero-shot initialization plus fast local tracking) as the most viable direction.

ClarissaApp | *Privacy-first personal health mobile app* Aug. 2025 – Feb. 2026

- Built a privacy-first mobile app for menstrual cycle tracking with React Native (Expo) and TypeScript.
- Implemented calendar-based cycle logging with symptom tracking and three themes (dark/light/red).
- Engineered offline-first storage with expo-sqlite, modular components/hooks, and a clear UX for fast daily logging.

Full-Stack ISP Comparison | *Check24 Scholarship Challenge* May 2025 – Jun. 2025

- Implemented a modular full-stack application (React + Vite + Tailwind frontend, Node.js/Express backend).
- Integrated timeout/health handling and an ISP provider normalization layer to produce consistent comparisons.
- Developed a share results feature via a TinyURL export persisting the results within the link.

Learning-Based Inverse Kinematics | *Bachelor Thesis* Jun. 2024 – Nov. 2024

- Designed and implemented learning-based IK solvers for 2- and 3-DOF manipulators in Python and PyTorch, comparing direct-angle regression (MLP and reinforcement learning with PPO) with probabilistic outputs (MLP, LSTM, hybrid MLP+LSTM).
- Built a synthetic data generation and evaluation pipeline; ran controlled experiments on a 10k-sample test set to quantify accuracy, convergence, and runtime across architectures and output parameterizations.
- Found that a simple MLP with direct-angle prediction consistently outperformed more complex models, highlighting a non-trivial complexity-performance trade-off and yielding clear guidance for practical IK model selection.

Ironman | *3.8km swim, 180km bike, 42.2km run; Overall Time : 14:21:26h* Oct. 2022 – Aug. 2024

- Completed a full Ironman triathlon through disciplined training and a goal-oriented race plan, demonstrating resilience and on-the-fly problem-solving under fatigue and variable race conditions to reach the finish.

Internet Lab 2 | *Advanced Practical Networking Course* Oct. 2023 – Mar. 2024

- Completed course labs on advanced networking topics in a two-person team, building strong proficiency in GNU/Linux CLI and Cisco IOS for complex network configurations.
- Co-created a full WebAuthn lab from scratch (lecture + prelab + hands-on) covering passkeys, FIDO2/WebAuthn flows, and security keys, enabling peers to configure WebAuthn end-to-end.
- Engineered the practical stack with Keycloak and a Spring Boot demo app, guiding students through server setup, WebAuthn policy configuration, and packet analysis to deepen protocol understanding.

Cyber-Physical Systems Survey | *Advanced Cyber-Physical Seminar* Apr. 2023 – Jul. 2023

- Produced an in-depth research survey of numerical inverse kinematics in cluttered environments for robots.
- Evaluated Jacobian-based, heuristic, and meta-heuristic methods, for convergence, robustness, and performance.

FreshFinder | hackaTUM - HelloFresh Challenge Nov. 2023

- Developed a React + TypeScript web app for collaborative, parallel cooking sessions with a team of 4
- Designed a recipe dependency-graph architecture for task distribution using Supabase Realtime.
- Integrated the OpenAI API to dynamically adapt recipes for individual preferences and amounts.

StreamFindA | hackaTUM - Burda Challenge Nov. 2022

- Built a React front end with a Django/Python backend for swipe-based movie/series recommendations.
- Implemented a recommendation algorithm based on pairwise choices using dwell time and negative feedback.
- Parsed and normalized the provided multi-platform catalog to a usable format to power reliable recommendations.

Computer Architecture Practical Course | *Advanced Practical Course* Apr. 2021 – Jul. 2021

- Implemented iterative space-filling Z-curve generators in C and Assembly with SVG export and performance benchmarks.

Humanoid-controlled robot arm | *Student Engineering Academy* Sep. 2018 – Jul. 2019

- Programmed a self-built robot arm controlled via a humanoid interface.
- Developed a virtual movement visualization of the arm sleeve with flex sensors in Blender with Python.
- Programmed Arduino microcontroller in C/C++ to read movement data from the arm sleeve, transmit them wirelessly via radio using the SPI protocol to the robot arm and map them to the servo motors.

MATLAB physics visualizations | *International Summer Physics Institute, University of Notre Dame* Jul. 2018

- Completed Cosmic Ray Detection team project by building a detector plus analyzing and visualizing the data.
- Visualized and analyzed CERN CMS particle-physics data using MATLAB.

Technical Skills

Languages: C#, Java, TypeScript, JavaScript, Python, C/C++, SQL, MATLAB | **Standards**: OpenDRIVE, OpenSCENARIO
Frameworks: React, React Native, Unity, PyTorch, Eclipse RCP, EMF, JavaFX, Supabase/PostgreSQL, WebRTC
Developer Skills: Git, Linux, GitHub Actions, Jenkins, Conan, SWTBot, JUnit, Playwright, Vitest, Rider, VS Code, VIM Motions, Jira

Further Highschool Experience

STEM Programs and Workshops 2017 – 2019

- MINT-EC Workshop: Masterclass Particle Physics (Jun. 2019).
- Fraunhofer Talent Take Off - Vernetzen (2019).
- Regional MINT-Camp "Life on Mars" (May–Jun. 2019): Developed concepts for life-essential resources.
- Tag der Kombinatorik — group-based local math contest (Feb. 2019).
- Student Engineer Academy — project track (2018–2019).
- Fraunhofer Talent School (Mar. 2018): Implemented an image classifier (cats vs. dogs).
- Student Engineer Academy — Programming in C (2017–2018).

Early student Internships 2015 – 2019

- Bosch, Stuttgart-Feuerbach — voluntary engineering internship (Apr., 2019).
- Palfinger, Ilsfeld — voluntary mechanic internship (Sep. – Oct., 2017).
- Getrag, Untergruppenbach — voluntary mechanic internship (Sep., 2017).
- Vector, Stuttgart — "Mitmachen Ehrensache" social one-day placement (Dec., 2015).
- Eberhard Schweizer Grundschule Ganztagesgruppe — social week-long placement in nursery school (Nov., 2015).

Music and Community 2009 – 2020

- Leadership: Youth spokesman, Heinriet Sports Club (2016–2020); school mediator for grades 5–6 (2016–2020).
- Awards: D2 (2017, excellent–very good); Jugend musiziert 2nd prize (2015); D1 (2014, very good); scholarship (2013) and gifted promotion (2012) in trombone.
- Piano and trombone (advanced); District youth orchestra Heilbronn (2017–2020); Musikverein Heinriet (2014–2020); school band 1st trombone (2012–2014).
- Theater: Castle theater productions as trombonist and actor (2017: Das kleine Gespenst; 2016: Some like it hot).